Operations Notes

Deployment checks and incident fixes

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Session 21 — Final DevOps Capstone

Project overview

Create, list, read, edit and delete deployment notes.

A responsive browser frontend calls a FastAPI backend.

PostgreSQL stores notes; Alembic manages the schema.

The DevOps workflow tests, scans, releases and monitors the app.

Tech stack

Application: HTML / JavaScript, Vite, NGINX, Python and FastAPI.

Data and tests: PostgreSQL, SQLAlchemy, Alembic and pytest.

Delivery: GitHub Actions, Docker, GHCR, Helm and Argo CD.

Infrastructure and metrics: Terraform, AWS EKS, Prometheus and Grafana.

Application architecture

Browser → NGINX Ingress → Frontend Vite + NGINX → Backend FastAPI → PostgreSQL Alembic

/ → frontend /api → backend

Backend /metrics → Prometheus → Grafana

Local: Docker Compose Cluster: Helm + Argo CD

API and database

GET /health checks the process; /ready checks the database.

GET, POST, PUT and DELETE implement the notes API.

Migration 0001_notes creates the notes table and records its revision.

A database PVC preserves notes across application restarts.

Testing and security

23 tests pass with SQLite and dedicated PostgreSQL test storage.

Tests cover CRUD, validation, readiness, metrics and migrations.

Bandit, pip-audit and full-history Gitleaks provide quality gates.

Trivy fails the pipeline on HIGH or CRITICAL findings in either image.

CI/CD flow

Push to main → backend tests and frontend build.

Security gates → backend and frontend Docker builds.

Trivy passes → both images pushed to GHCR with the full commit SHA.

Helm deploys to kind; CI verifies PostgreSQL and two replicas per tier.

AWS infrastructure with Terraform

Selected Region: ap-southeast-2; CLI profile: devops-homework.

Terraform defines a VPC, two public subnets and a managed EKS node group.

Provisioned 5 October; live EKS and access update verified 13 October 2026.

Verify the ACTIVE Free plan and credits; capture destroy completion after evaluation.

Kubernetes and Helm

The application namespace is project-three-tier.

Frontend and backend each run at least two replicas.

ClusterIP Services connect the tiers; PostgreSQL uses a StatefulSet and PVC.

NGINX Ingress routes / and /api; EKS hostname: notes.aws.project.local.

GitOps with Argo CD

Argo CD tracks the homework repository main branch.

It renders the Helm chart with committed EKS or local values.

Automatic sync and self-healing reconcile the application.

The previous Flux runtime is paused to prevent competing writers.

Prometheus and Grafana

GET /metrics exposes request counts, errors, latency, CPU and memory.

Prometheus discovers and scrapes each backend Pod every five seconds.

Grafana displays application metrics from the Prometheus datasource.

Generate traffic, then show UP targets and populated dashboard panels.

Key learnings

Verify API behavior and stored data, not only the homepage.

Use readiness probes and immutable image tags to identify releases.

Keep credentials private and distinguish old evidence from current execution.

Treat AWS cleanup as part of delivery; record the complete live walkthrough.